



TRINITY GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT



YEAR 11 2008 ASSESSMENT TASK 4

MATHEMATICS
YEARLY EXAMINATION

Tuesday, 26th August, 2008
8:30 am to 11:00 am

Time Allowed – *Two and a half hours plus 5 minutes reading*

WEIGHTING 40% towards final assessment

Outcomes referred to: P1, P2, P3, P4, P5, P6, P7.

INSTRUCTIONS:

1. Attempt ALL questions.
2. Show all necessary working in every question.
3. Begin each question on a separate piece of paper.
4. There are 10 questions worth 12 marks each. Total marks 120.
5. Marks for individual part questions are shown beside each part.
6. Non-programmable silent Board of Studies approved calculators are permitted.
7. Write your name, class and your teacher's name on the front of each answer page.

Question 1 (12 marks)

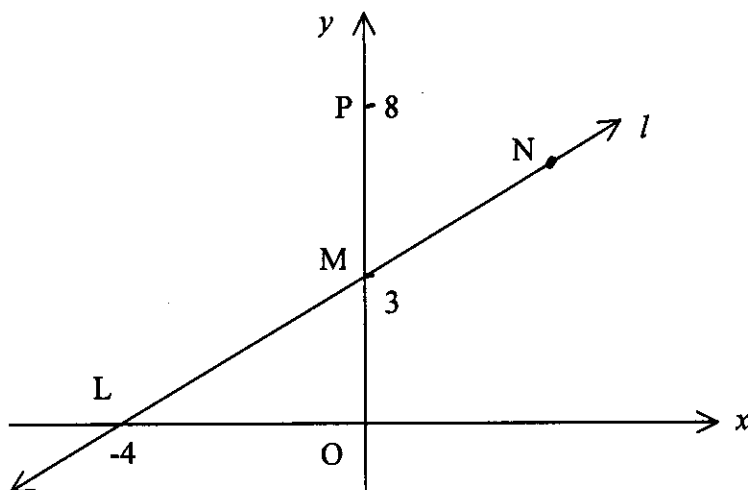
Start each new question on a new page. Write your name and your teacher's name at the top of the page

	Marks
a) Find the value of $13^{-1.2}$ correct to two significant figures.	1
b) Factorise $2x^2 + 3x - 2$.	2
c) Rationalise the denominator of $\frac{4}{\sqrt{5} + 2}$.	2
d) Write 0.000 581 in scientific notation.	1
e) Simplify $3\sqrt{32}$.	2
f) The price of a car increased by 4%. If the car now costs \$15 080, how much did it cost previously?	2
g) If $A = P(1 + \frac{r}{100})^n$, find P when $A = 352$, $r = 12$ and $n = 5$.	2

Question 2 (12 marks)

Start each new question on a new page. Write your name and your teacher's name at the top of the page.

Marks



The line l cuts the x axis at $L(-4,0)$ and y axis at $M(0,3)$ as shown. N is a point on the line l , and P is the point $(0,8)$.

Copy the diagram into your Writing Booklet.

- | | | |
|----|---|---|
| a) | Find the equation of the line l . | 2 |
| b) | Show that the point $(16,15)$ lies on the line l . | 1 |
| c) | By considering the lengths of ML and MP , show that $\triangle LMP$ is isosceles. | 2 |
| d) | Calculate the gradient of the line PL . | 1 |
| e) | M is the midpoint of the interval LN . Find the coordinates of the point N . | 2 |
| f) | Show that $\angle NPL$ is a right angle. | 2 |
| g) | Find the equation of the circle that passes through the points N , P , and L . | 2 |

Question 3 (12 marks)

Start each new question on a new page. Write your name and your teacher's name at the top of the page.

Marks

a) Differentiate $y = 5x^2 - 3x$ from first principles.

3

b) Differentiate:

(i) $7x^6 + \frac{2}{x^3}$

2

(ii) $6x^2\sqrt{x}$

2

(iii) $\frac{(2x-9)^3}{5x+1}$

2

c) Find the equation of the tangent to the curve $y = 2x^3 - 7$ at the point where $x = 1$.

3

Question 4 (12 marks)

Start each new question on a new page. Write your name and your teacher's name at the top of the page.

	Marks
a) Simplify	
(i) $6a - 3b - a + 7b$	1
(ii) $(-4t^2)^3$	1
b) Factorise fully:	
(i) $x^2 - 2x - 24$	2
(ii) $5y - 15 + xy - 3x$	1
c) Expand and simplify: $(4x - 3)^2$	2
d) Simplify the following: $\frac{x-1}{2} - \frac{x+4}{3}$	2
e) Simplify: $\frac{5m+10}{m^2-m-2} \div \frac{m^2-4}{3m+3}$	3

Question 5 (12 marks)

Start each new question on a new page. Write your name and your teacher's name at the top of the page.

Marks

a) Solve the following:

(i) $\frac{x+3}{7} + \frac{x-1}{3} = 4$

2

(iii) $2y^2 + y - 3 = 0$

2

b) Solve the following and graph your solution on a number line.

2

$$|3x + 5| > 22$$

c) Solve the following giving your answer correct to 2 decimal places.

2

$$x^2 + 2x - 7 = 0$$

d) Solve the following pairs of simultaneous equations.

2

(i)
$$\begin{aligned} 5x - y &= 19 \\ 2x + 5y &= -14 \end{aligned}$$

(ii)
$$\begin{aligned} x - y - 7 &= 0 \\ xy &= -12 \end{aligned}$$

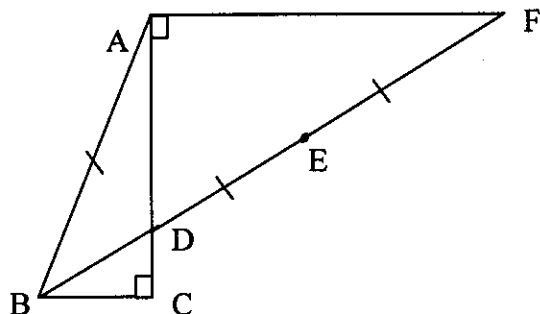
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Question 6 (12 marks)

Start each new question on a new page. Write your name and your teacher's name at the top of the page.

Marks

a)



In the diagram, $AC \perp BC$, $AC \perp AF$ and $AB = DE = EF$.

Copy or trace the diagram on to your answer page.

- | | | |
|-------|---|---|
| (i) | Show that $\angle DBC = \angle DFA$. | 2 |
| (ii) | On your diagram, mark the point G on the line AF such that $EG \parallel AC$. Show that $\triangle AGE \equiv \triangle FGE$. | 2 |
| (iii) | Prove that $\angle ABD = 2\angle DBC$. | 2 |
-
- b) A stick 1 metre long placed vertically on the ground casts a shadow 60 cm long, and at the same time a building casts a shadow 3 m long. What is the height of the building? 2
- c) The height of a cone is twice its radius. Find the volume of the cone in terms of r . 2
- d) To cover a rectangular bench top, 400 tiles, each 10 cm by 5 cm are used. If square tiles of length 4 cm are used, how many would be needed? 2

Question 7 (12 marks)

Start each new question on a new page. Write your name and your teacher's name at the top of the page.

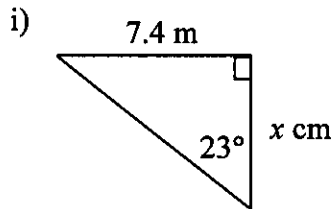
	Marks
a) If $f(x) = x^3 - 6x^2 + 5$ find $f(-2)$.	1
b) Given $f(x) = x^2 - 4$, find values of x when $f(x) = 32$.	1
c) Show whether $f(x) = 4x - x^3$ is an odd or even function.	2
d) Sketch the region which satisfies the following conditions, $x \geq -3$, $y < 2x + 1$ and $x^2 + y^2 \geq 9$, showing x and y intercepts and asymptotes.	3
e) State the domain and range of the curves below:	
(i) $y = \frac{1}{x+3}$	1
(ii) $y = \sqrt{4-x^2}$	1
f) Sketch the curve $y = x^2 - 3x - 4$ showing the axis of symmetry, the x and y intercepts and the vertex.	3

Question 8 (12 marks)

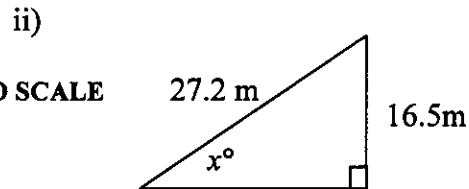
Start each new question on a new page. Write your name and your teacher's name at the to

Marks

- a) Find the value of x in the following diagrams, correct to the nearest whole number.



FIGURES NOT TO SCALE



4

- b) (i) Find θ to the nearest minute if $\sin \theta = 0.82$.

1

- (ii) If $\cos \theta = \frac{2}{7}$, find the exact ratio of $\tan \theta$.

2

- c) The angle of depression from the top of an 18 m high building to a person on the ground below is $47^\circ 28'$. How far out from the building is the person, to 1 decimal place.

2

- d) A plane flies south from Sydney for 590 km and turns and flies west for 360 km.

- (i) Draw a diagram for this information.

1

- (iii) What is its bearing from Sydney to the nearest degree?

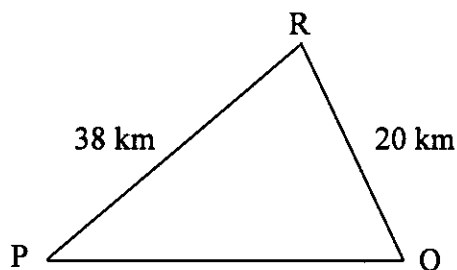
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Question 9 (12 marks)

Start each new question on a new page. Write your name and your teacher's name at the top.

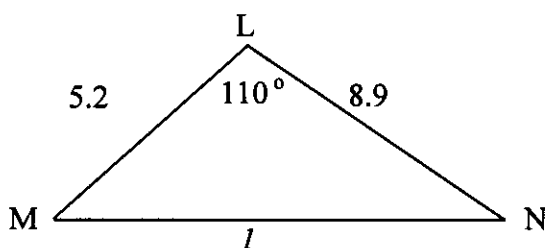
Marks

- a) In the diagram, the point Q is due east of P. The point R is 38km from P and 20km from Q. The bearing of R from Q is 325° .



- (i) What is the size of $\angle PQR$? 2
- (ii) What is the bearing of R from P? 1
- b) Find the exact value of $5 \sin 30^\circ \sec 30^\circ$ and rationalise the denominator. 2

c)



In the diagram, LMN is a triangle where $LM = 5.2$ metres, $LN = 8.9$ metres and angle $MLN = 110^\circ$.

- (i) Find the value of l correct to 1 decimal place. 2
- (ii) Calculate the area of triangle of triangle LMN. 1
- d) Simplify $\frac{\tan \theta \sec \theta}{1 + \tan^2 \theta}$ 2
- e) Solve the equation $2 \sin \theta - 1 = 0$ for $0^\circ \leq \theta \leq 360^\circ$. 2

Question 10 (12 marks)

Start each new question on a new page. Write your name and your teacher's name at the top of the page.

	Marks
a) If α and β are the roots of the equation $x^2 + 5x - 8 = 0$, find the value of :	
(i) $\alpha + \beta$	1
(ii) $\alpha\beta$	1
(iii) $(\alpha - 2)(\beta - 2)$	2
(iv) $\alpha^2 + \beta^2$	2
b) (i) Find the discriminant of the quadratic equation $x^2 - 2x + 3$.	1
(ii) What does your answer in part (i) indicate about the roots of the quadratic $x^2 - 2x + 3$?	1
c) Solve $4x - x^2 \geq 0$.	2
d) Find the values of the constants A and B such that $20x - 17 \equiv A(x - 4) - B(5x + 1)$	2

End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question 1

a) $13^{-1.2} = 0.046$

1

b) $(2x-1)(x+2)$

2

c) $\frac{4}{\sqrt{5}+2} \times \frac{\sqrt{5}-2}{\sqrt{5}-2} = \frac{4\sqrt{5}-8}{5-4} = 4\sqrt{5}-8$

2

d) 5.81×10^4

1

e) $3\sqrt{32} = 3 \times \sqrt{16} \times \sqrt{2}$
 $= 3 \times 4 \times \sqrt{2}$
 $= \underline{12\sqrt{2}}$

2

f) $1.04x = 15080$
 $x = 15080 \div 1.04$
 $= \underline{\$14500}$

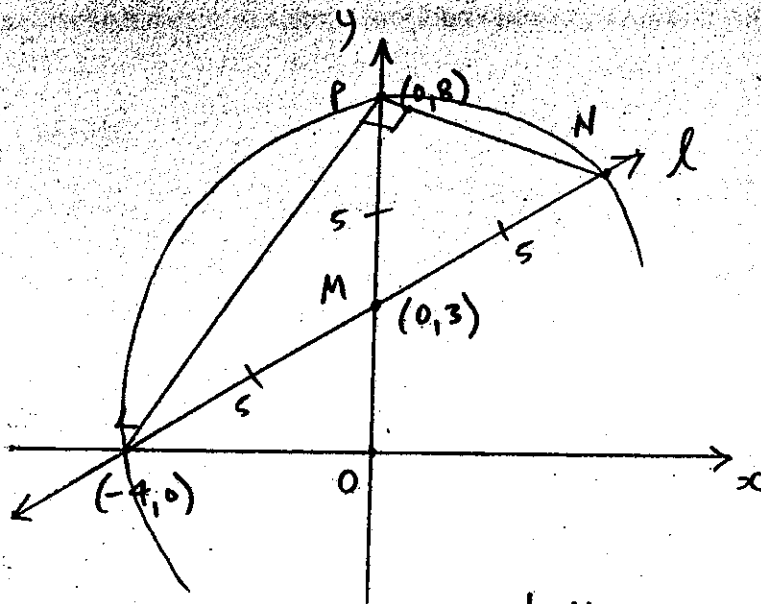
2

g) $352 = P(1+0.12)^5$

$P = \frac{352}{1.12^5} = 199.73 \quad (2 \text{ dp})$

2

Question 2 - Solutions (DK)



$$(a) \quad m_l = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\begin{matrix} x_1 & y_1 & x_2 & y_2 \\ (-4, 0) & (0, 3) \end{matrix} \quad = \frac{3-0}{0-(-4)}$$

Alternative:
two point formula) $= \frac{3}{4} \quad (1)$

$$y\text{-intercept: } b = 3$$

$$\therefore \text{equation: } y = \frac{3}{4}x + 3 \quad (1)$$

b) Subst. (16, 15) into $y = \frac{3}{4}x + 3$.

$$\text{RHS} = \frac{3}{4}(16) + 3$$

$$= 12 + 3$$

$$= 15$$

$$= \text{LHS}$$

$$= y.$$

$\therefore (16, 15)$ lies on the line l . (1)

$$ML = \sqrt{(0-(-4))^2 + (3-0)^2} = \sqrt{16+9} = 5 \quad (1)$$

$$MP = 8-3 = 5$$

since $ML = MP = 5$ then $\triangle LMP$ is isosceles. (1)

$$(d) \quad m_{PL} = \frac{8-0}{0-(-4)} = \frac{8}{4} = 2. \quad (1)$$

$$(e) \quad L(-4, 0) \quad N(x_2, y_2)$$

$$M\left(\frac{-4+x_2}{2}, \frac{0+y_2}{2}\right) = M(0, 3)$$

$$\therefore \frac{-4+x_2}{2} = 0 \quad \text{and} \quad \frac{0+y_2}{2} = 3$$

$$-4+x_2 = 0$$

$$y_2 = 6$$

$$x_2 = 4.$$

$$\therefore N(4, 6) \quad (2)$$

(f) From (d), $m_{PL} = 2$. (Alternative: Pythagoras Thm)

$$\begin{matrix} x_1 & y_1 & x_2 & y_2 \\ P(0, 8) & N(4, 6) \end{matrix} \quad m_{PN} = \frac{6-8}{4-0} = \frac{-2}{4} = -\frac{1}{2} \quad (1)$$

$$\text{Now, } m_{PL} \times m_{PN} = 2 \times -\frac{1}{2} = -1 \quad (1)$$

$$\therefore \angle P \perp PN$$

$$\& \quad \angle NPL = 90^\circ.$$

$$(g) \quad \begin{matrix} \text{Centre: } (0, 3) \\ \text{radius} = ML = MN = MP = 5 \end{matrix} \quad (1)$$

$$\therefore x^2 + (y-3)^2 = 25 \quad (1)$$

Question 3

a) $f(x) = 5x^2 - 3x$

$$\begin{aligned} f(x+h) &= 5(x+h)^2 - 3(x+h) \\ &= 5(x^2 + 2xh + h^2) - 3x - 3h \\ &= 5x^2 + 10xh + 5h^2 - 3x - 3h \end{aligned}$$

$$\begin{aligned} \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{5x^2 + 10xh + 5h^2 - 3x - 3h - 5x^2 + 3x}{h} \\ &= \lim_{h \rightarrow 0} \frac{10xh + 5h^2 - 3h}{h} \\ &= \lim_{h \rightarrow 0} 10x - 5h - 3 \\ &= 10x - 3 \end{aligned}$$

b) (i) $42x^5 - 6x^{-4} = 42x^5 - \frac{6}{x^4}$

(ii) $6x^2 \times x^{\frac{1}{2}} = 6x^{\frac{5}{2}} \quad \frac{d}{dx} (6x^{\frac{5}{2}}) = \underline{\underline{15x^{\frac{3}{2}}}}$

(iii)
$$\begin{aligned} &\frac{(5x+1) \times 3 \times 2(2x-9)^2 - (2x-9)^3 \times 5}{(5x+1)^2} \\ &= \frac{(2x-9)^2 [30x+6 - 5(2x-9)]}{(5x+1)^2} \\ &= \frac{(2x-9)^2 (30x+6 - 10x+45)}{(5x+1)^2} \\ &= \frac{(2x-9)^2 (20x+51)}{(5x+1)^2} \end{aligned}$$

$$c) \quad y = 2x^3 - 7, \quad y_1 = 2(1)^3 - 7 = 2 - 7 = -5$$

$$\frac{dy}{dx} = 6x^2 = 6(1)^2 = 6$$

$$y - y_1 = m(x - x_1)$$

$$y - (-5) = 6(x - 1)$$

$$y + 5 = 6x + 6$$

$$\underline{y = 6x + 1}$$

$$\text{or } \underline{6x - y + 1 = 0}$$

Question 4

a) (i) $5a + 4b$

1

(ii) $-64t^6$

1

2

b) (i) $(x-6)(x+4)$

(ii) $5(y-3) + x(y-3) = (5+x)(y-3)$

1

c) $(4x-3)^2 = 16x^2 - 24x + 9$

2

d) $\frac{x-1}{2} - \frac{x+4}{3} = \frac{3(x-1)}{6} - \frac{2(x+4)}{6}$

$$= \frac{3x-3-2x-8}{6}$$

$$= \frac{x-11}{6}$$

2

e) $\frac{5m+10}{m^2-m-2} \div \frac{x^2-4}{3m+3} = \frac{5(m+2)}{(m-2)(m+1)} \times \frac{3(m+1)}{(m+2)(m-2)}$

$$= \frac{5}{(m-2)^2}$$

3

Question 5

a) (i) $\frac{x+3}{7} + \frac{x-1}{3} = 4$

$$3x+9 + 7x-7 = 84 \quad (1)$$

$$10x+2 = 84$$

$$10x = 82$$

$$x = 8\frac{1}{5} \quad (1)$$

(ii) $2y^2 - 2y + 3y - 3 = 0$

$$2y(y-1) + 3(y-1) = 0 \quad (1)$$

$$(2y+3)(y-1) = 0$$

$$2y+3=0, y-1=0$$

$$2y = -3$$

$$y = -\frac{3}{2} \text{ or } y=1 \quad (1)$$

b) $|3x+5| > 22$

$$3x+5 > 22$$

$$3x > 17$$

$$x > \frac{17}{3}$$

$$x > 5\frac{2}{3} \quad (1)$$

$$-(3x+5) > 22$$

$$-3x-5 > 22$$

$$-3x > 27$$

$$x < -\frac{27}{3}$$

$$x < -9 \quad (1)$$

c) $x = \frac{-2 \pm \sqrt{2^2 - 4 \times 1 \times -7}}{2 \times 1}$

$$= \frac{-2 \pm \sqrt{4 + 28}}{2}$$

$$= \frac{-2 \pm \sqrt{32}}{2}$$

$$(1) \quad \frac{2}{2} \quad (1)$$

$$= 1.83 \text{ or } -3.83$$

$$d) (i) \quad 5x - y = 19 \quad (1)$$

$$2x + 5y = -14 \quad (2)$$

$$(1) \times 5 \quad 25x - 5y = 95 \quad (3)$$

$$(2) + (3) \quad 27x = 81 \quad (1)$$

$$x = 3$$

Sub $x=3$ into (1)

$$5(3) - y = 19$$

$$15 - y = 19$$

$$-y = 4$$

$$y = -4 \quad (1)$$

$$\therefore \underline{x=3, y=-4}$$

$$(ii) \quad x - y - 7 = 0 \quad (1)$$

$$xy = -12 \quad (2)$$

$$\text{From (1)} \quad y = x - 7 \quad (3)$$

Sub $y = x - 7$ into (2)

$$x(x-7) = -12$$

$$x^2 - 7x + 12 = 0$$

$$(x-4)(x-3) = 0$$

$$x = 3, 4$$

$$\text{If } x=3, y = 3-7 = -4$$

$$\text{If } x=4, y = 4-7 = -3$$

$$\therefore \underline{x=3, y=-4 \text{ and } x=4, y=-3}$$

(1)

(1)

Question 6

a) (i) $\angle BCD = \angle FAC = 90^\circ$ (given)

$\therefore AF \parallel BC$ (alternate angles equal only on $\parallel$)

$\therefore \angle OBC = \angle OFA$ (alternate angles in parallel lines)

(ii) $\angle AGE = 90^\circ$ (co-int with $\angle GAO$, $AO \parallel GE$)

$\angle EGF = 90^\circ$ ($\angle$ on straight line)

$\therefore \angle AGE = \angle EGF$ (both 90°)

Since E is midpoint of DF, G must be midpoint of AF (equal intercept theorem)

$\therefore AG = GF$

$GE = GE$ (common)

$\therefore \triangle AGE \cong \triangle FGE$ (SAS)

① two reasons

① third reason
+ congruence

(iii) Let $\angle OBC = \alpha$

$\angle AFE = \alpha$ from (i)

$\angle EAF = \angle AFE$ (corresponding angles
 $\triangle AGE \cong \triangle FGE$)

$= \alpha$

$\therefore \angle AEO = 2\alpha$ (ext angle of $\triangle$ equals sum of int opp angles)

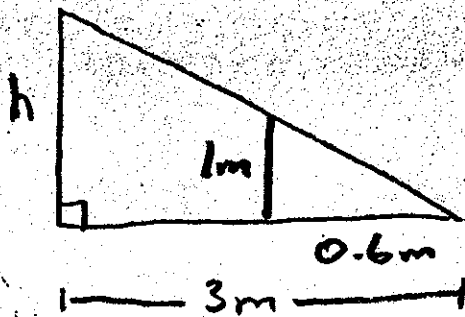
Now, $AE = EF$ (corresponding sides
 $\triangle AGE \cong \triangle FGE$)

$\therefore AB = AE$ both equal to EF

$\angle ABC = \angle AEO$ (base angles of isosceles)
 $= 2\alpha = 2\angle OBC$

or proof
with
angle sum
of $\triangle$

b)



$$\frac{h}{3} = \frac{1}{0.6}$$

$$h = \frac{1}{0.6} \times 3$$

$$= \underline{5m}$$

c)

$$V = \frac{1}{3} \pi r^2 h$$

$$V = \pi r^2 \times 2r \times \frac{1}{3}$$

$$= \underline{2\pi r^3 \times \frac{1}{3}} = \frac{2}{3} \pi r^3$$

(Core volume = $\frac{1}{3} \pi r^2 h$)

d)

$$\text{Total area} = (10 \times 5) \times 400$$

$$= 20000 \text{ cm}^2$$

$$\text{N. of tiles} = 20000 \div (4 \times 4)$$

$$= \underline{1250}$$

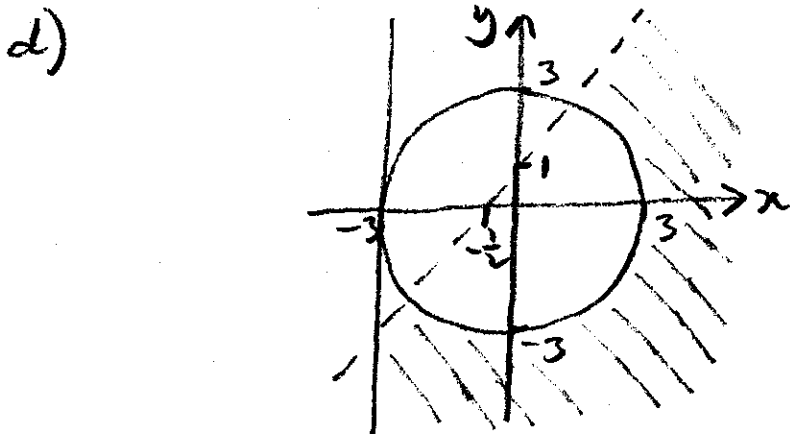
Question 7

a) $f(-2) = (-2)^3 - 6(-2)^2 + 5 = \underline{-27}$ 1

b) $x^2 - 4 = 32$
 $x^2 = 36$
 $x = \pm 6$ 1

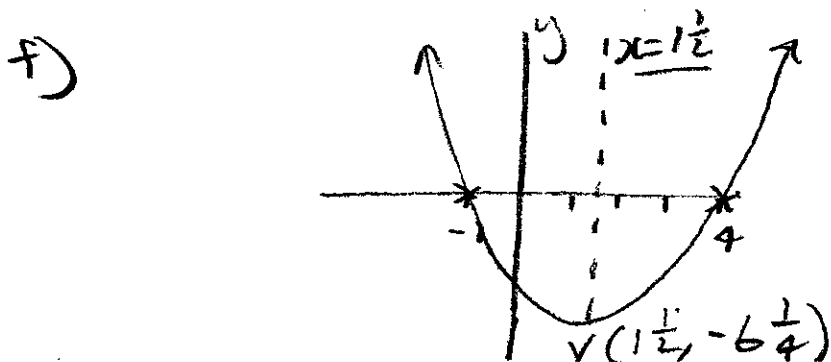
c) $f(-x) = 4(-x) - (-x)^3$
 $= -4x + x^3$
 $= -f(x)$

$\therefore f(x)$ is odd 2



e) (i) Domain : all real x , $x \neq -3$
Range : all real y , $y \neq 0$ 1

(ii) Domain : $-2 \leq x \leq 2$
Range : $0 \leq y \leq 2$ 1



Question 8

a) (i) $\tan 23 = \frac{7.4}{x}$

$$x = 7.4 \div \tan 23$$
$$= \underline{17\text{m}} \quad (2)$$

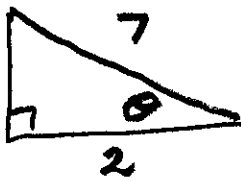
(ii) $\sin x = \frac{16.5}{27.2}$

$$x = \underline{37^\circ} \quad (2)$$

b) (i) $\sin \theta = 0.82$

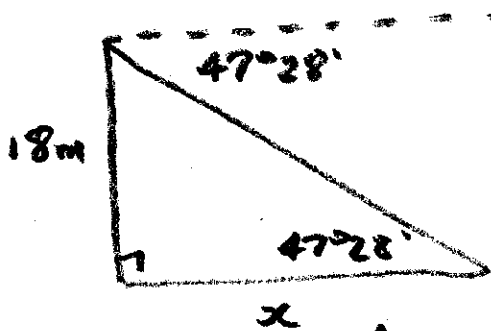
$$\theta = \underline{55^\circ 5'} \quad (1)$$

(ii) $\sqrt{45}$



$$\tan \theta = \frac{\sqrt{45}}{2} \quad (2)$$

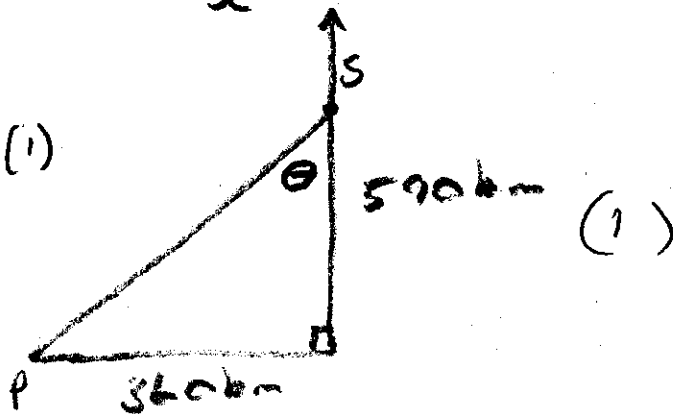
c)



$$\tan 47^\circ 28' = \frac{18}{x}$$

$$x = \frac{18}{\tan 47^\circ 28'}$$
$$= \underline{16.5\text{m}} \quad (2)$$

d) (i)



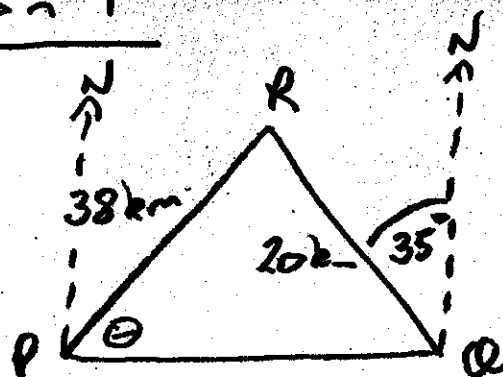
(ii) $\tan \theta = \frac{360}{590}$

$$\theta = 31^\circ 23'$$

$$\text{Bearing} = 180 + 31^\circ 23'$$
$$= \underline{211^\circ 23'} \quad (2)$$

Question 9

a)



$$(i) \angle PQR = 90 - 35 = \underline{55^\circ} \quad (2)$$

$$(ii) \frac{\sin Q}{20} = \frac{\sin 55}{38}$$

$$\sin Q = \frac{\sin 55}{38} \times 20$$

$$\angle Q = 25^\circ 32'$$

$$\therefore \text{Bearing is } 90 - 25^\circ 32' = \underline{64^\circ 28'} \quad (1)$$

$$b) 5 \sin 30 \sec 30 = 5 \times \frac{1}{2} \times \frac{1}{\frac{\sqrt{3}}{2}}$$

$$= \frac{5}{2} \times \frac{2}{\sqrt{3}} \quad (M1)$$

$$= \frac{5}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$= \frac{5\sqrt{3}}{3} \quad (1)$$

$$c) (i) L^2 = 5.2^2 + 8.9^2 - 2 \times 5.2 \times 8.9 \times \cos 110 \quad (M1)$$

$$= \underline{11.7 \text{ m}} \quad (1)$$

$$(ii) A = \frac{1}{2} \times 5.2 \times 8.9 \times \sin 110$$

$$= \underline{21.7 \text{ m}^2} \quad (1)$$

$$\begin{aligned}
 d) \quad \frac{\tan \theta \sec \theta}{1 + \tan^2 \theta} &= \frac{\tan \theta \sec \theta}{\sec^2 \theta} \\
 &= \frac{\tan \theta}{\sec \theta} \quad (M1) \\
 &= \frac{\sin \theta}{\cos \theta} \div \frac{1}{\cos \theta} \\
 &= \sin \theta \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 e) \quad 2 \sin \theta - 1 &= 0 \\
 \sin \theta &= \frac{1}{2}
 \end{aligned}$$

S	A
T	C

$$\begin{aligned}
 \theta &= 30^\circ, 180 - 30 \\
 &= 30^\circ, 150^\circ \\
 &\quad \text{---} \quad (1)
 \end{aligned}$$

Question 10

a) $x^2 + 5x - 8 = 0$

(i) $\alpha + \beta = \frac{-b}{a} = -5$ 1

(ii) $\alpha\beta = \frac{c}{a} = -8$ 1

(iii) $\alpha\beta - 2\alpha - 2\beta + 4 = \alpha\beta - 2(\alpha + \beta) + 4$
 $= -8 - 2(-5) + 4$
 $= -8 + 10 + 4$
 $= 6$ 2

(iv) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
 $= (-5)^2 - 2(-8)$
 $= 25 + 16$
 $= 41$ 2

b) (i) $\Delta = (-2)^2 - 4 \times 1 \times 3$
 $= 4 - 12$
 $= -8$ 1

(ii) No real roots since $\Delta < 0$ 1

c) $4x - x^2 \geq 0$
 $x(4 - x) \geq 0$ 2

$$\begin{array}{c} \hline 0 \quad 4 \\ \hline \end{array}$$

$0 \leq x \leq 4$

$$d) \quad 20x - 17 \equiv Ax - 4A - 5Bx - B \\ \equiv (A - 5B)x - (4A + B)$$

$$A - 5B = 20 \quad (1)$$

$$4A + B = 17 \quad (2)$$

$$+ \times (1) \quad 4A - 20B = 80 \quad (3)$$

$$(3) - (2) \quad -21B = 63$$

$$\underline{B = -3}$$

$$A + 15 = 20$$

$$\underline{A = 5}$$

2